

SpaceYZ's IOD Mission Overview

SpaceYZ needs to conduct an In-Orbit Demonstration (IOD) of their latest product. This product is a new imaging camera equipped with AI capabilities, with specific parameters:

- 1U form factor (camera dimension)
- Operating voltage: 5V
- Power consumption: 3W
- Mass: 0.5kg

CTO's Request for Proposal

Your company needs to manage the IOD, handling all aspects of the mission, including:

- Platform selection (CubeSat and components)
- Mission-dedicated personnel
- Facility and testing
- Launch opportunity
- Ground station operations

Your Proposal

SpaceYZ requests a detailed PDF (2-3 pages). Your proposal should cover the following areas:

- CubeSat Platform Selection: Detail the specifications, size, and additional components of the proposed CubeSat platform, emphasizing reliability, compatibility, and cost-effectiveness.
- Mission Personnel: Outline the roles and expertise of project managers, engineers, technicians, and other relevant staff members assigned to the mission.
- Facility and Testing Procedures: Describe the assembly, integration, and testing processes, along with any specialized equipment or testing protocols.
- Launch Strategy: Present a strategy for securing a suitable launch opportunity, including potential providers, vehicle options, timelines, and contingency plans.
- Ground Station Operations: Detail the approach to ground station operations, including communication protocols, data transfer methods, and backup systems.

Additional Requirements

- Mission Name and Objectives.

Mission Name:	<i>Give a name to the CubeSat mission</i>
Mission Objective 1 (O1):	<i>Define the first objective of the mission</i>
Mission Objective 2 (O2):	<i>Define the second objective of the mission</i>
Mission Objective 3 (O3):	<i>Define the third objective of the mission</i>

- Times, Deadlines, Milestones: Provide clear timelines, milestones, and deliverables for each phase of the project.
- Payment Plans: Specify payment terms, schedules, upfront costs, and milestone payments.
- Project Costs: Include a detailed breakdown of project expenses, covering hardware, personnel, testing, launch services, and ground operations.
- Optional: Include conceptual designs or renderings of the CubeSat and its components for visual illustration.